

Md Bodrud-Doza, PhD

Postdoctoral Scholar | Department of Geography, Environment and Geomatics | University of Guelph

Resilient Agricultural Landscapes and Sustainable Food Production

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ACADEMIC PROFILE

Environmental scientist and geographer specializing in agricultural water quality, precision conservation, watershed modelling, and geospatial decision support. My research integrates environmental monitoring, GIS, machine learning, and process-based models to plan and evaluate conservation across agricultural landscapes. Current postdoctoral work extends this foundation toward adaptive environmental decision support, Earth observation and GeoAI, and food-system resilience. I also bring substantial experience in climate adaptation, science-policy practice, partnership development, and research proposal development across academic, government, NGO, and private-sector settings.

RESEARCH AREAS

Agricultural water quality | Precision conservation and conservation planning | Watershed modelling | Environmental monitoring | GIS and geospatial analysis | Machine learning | Environmental decision support | Climate resilience | Earth observation and GeoAI (emerging) | Food-system resilience (developing)

EDUCATION

PhD in Geography

September 2022 - August 2026

University of Guelph | Guelph, Ontario, Canada

Thesis: Advancing Precision Conservation in Tile-Drained Agricultural Watersheds through Integrated Geospatial Analysis, Machine Learning, and Process-Based Modeling.

Research focus: precision conservation, agricultural water quality, geospatial analysis, machine learning, and process-based watershed modelling.

MSc in Environmental Sciences

2014

Jahangirnagar University | Bangladesh

Thesis: An integrated approach for groundwater pollution evaluation of Dhaka City using multivariate statistical analysis, groundwater pollution indices, geostatistical modeling, and human health risk assessment.

BSc (Honours) in Environmental Sciences

2010 - 2013

Jahangirnagar University | Bangladesh

Project: Assessment of Groundwater Quality in Dhaka District Using GIS and Multivariate Statistical Techniques.

ACADEMIC AND RESEARCH APPOINTMENTS

Postdoctoral Scholar

September 2026 - Present

Department of Geography, Environment and Geomatics, University of Guelph | Guelph, Ontario, Canada

- Lead active Mitacs Accelerate research in the Healthy Lake Huron context; Ausable Bayfield Conservation Authority (ABCA) is the formal Mitacs partner.
- Integrate long-term watershed monitoring, process-based modelling, conservation and climate scenarios, environmental indicators, and practical decision support.
- Developing REAL Decision Lab, a research program focused on resilient agricultural landscapes and sustainable food production.

Graduate Research Assistant

May 2024 - August 2026

Department of Geography, Environment and Geomatics, University of Guelph | Guelph, Ontario, Canada

- Conducted distributed process-based watershed modelling to evaluate agricultural conservation practices in tile-drained landscapes.
- Integrated GIS, statistical methods, machine learning, and Agricultural Conservation Planning Framework (ACPF)-based planning for field-scale conservation and Great Lakes water-quality management.
- Collaborated with Environment and Climate Change Canada, Iowa State University, and academic partners.

Data Analyst

June 2024 - August 2024

Ausable Bayfield Conservation Authority | Ontario, Canada

- Applied statistical and machine-learning approaches to long-term water-quality data from a Lake Huron tributary.
- Supported watershed interpretation and development of monitoring and quality-assurance plans.

RESEARCH FUNDING, GRANTS, AND PROPOSAL DEVELOPMENT**Current and Awarded Research Support**

- Mitacs Accelerate (2026 - present) - active postdoctoral research supporting Healthy Lake Huron watershed monitoring, modelling, indicators, and conservation decision support; ABCA is the formal Mitacs partner.
- Environment and Climate Change Canada (2024 - 2026) - PhD project funding supporting agricultural watershed and precision-conservation research.

Submitted and Under Review

- Ontario Ministry of Agriculture, Food and Agribusiness (OMAFRA) - Field-Specific Precision Conservation for Cost-Effective Water Quality and Ecosystem Benefits in Ontario's Lake Erie Basin. Contributed to proposal development with lead applicant Dr. Prasad Daggupati; submitted and under review. Funding is not confirmed.
- Canada Postdoctoral Research Award (CPRA) program - submitted application for proposed food-system resilience and anticipatory decision-making research. No award or fellowship funding is confirmed.

Professional Proposal Development and Resource Mobilisation

- BRAC - contributed to and led proposal development that secured more than USD 10 million through 14 successful proposals; co-designed additional projects intended to mobilize up to USD 100 million and co-led BRAC's accreditation to the Green Climate Fund as a Direct Access Entity.
- CARE Bangladesh - co-authored a successful ecosystem-based adaptation proposal that secured €35 million from the UK Foreign, Commonwealth & Development Office (FCDO).

SELECTED AWARDS AND HONOURS

2025-2026	Ontario Graduate Scholarship (OGS), University of Guelph.
2025	Cecil H. Franklin Graduate Scholarship in Soil & Water Conservation, University of Guelph.
2025	Ted Symons Scholarship, University of Guelph.
2024, 2025	GEG Graduate Publication Award, University of Guelph.
2023	Arthur D. Latornell Graduate Scholarship, University of Guelph.
2023	Three Minute Thesis (3MT) College Heat Winner, College of Social and Applied Human Sciences, University of Guelph.
2022	International Doctoral Tuition Scholarship (IDTS), Special Dean's Award, and Kiyoko Miyanishi Graduate Geography Scholarship, University of Guelph.

PEER-REVIEWED PUBLICATIONS

Complete list as of October 2026: 60 peer-reviewed journal articles, including 10 first-authored articles. My name is bolded below.

First-Authored Articles

1. **Bodrud-Doza, M.**, Yang, W., Liu, Y., Yerubandi, R., McKague, K., James, D., DeVries, B., & Fraser, E. D. G. (2026). Operationalizing precision conservation: Aligning best management practice prioritization with local needs in agricultural watersheds. *International Soil and Water Conservation Research*, 100688. DOI: [10.1016/j.iswcr.2026.100688](https://doi.org/10.1016/j.iswcr.2026.100688)
2. **Bodrud-Doza, M.**, Yang, W., Liu, Y., Yerubandi, R., Thompson, M., McKague, K., James, D., DeVries, B., & Fraser, E. D. G. (2026). Advancing precision conservation through machine learning for field-level prioritization of agricultural best management practices. *International Soil and Water Conservation Research*, 100688. DOI: [10.1016/j.iswcr.2026.100688](https://doi.org/10.1016/j.iswcr.2026.100688)
3. **Bodrud-Doza, M.**, Yang, W., Liu, Y., Yerubandi, R., Daggupati, P., DeVries, B., & Fraser, E. D. G. (2025). Evaluating best management practices for nutrient load reductions in tile-drained watersheds of the Laurentian Great Lakes Basin: A literature review. *Science of The Total Environment*, 965, 178657. DOI: [10.1016/j.scitotenv.2025.178657](https://doi.org/10.1016/j.scitotenv.2025.178657)
4. **Bodrud-Doza, M.**, Yang, W., Miranda, R Q; Martin, A; DeVries, B; Fraser, E D G. (2023). Towards implementing precision conservation practices in agricultural watersheds: A review of the use and prospects of Spatial Decision Support Tools and Systems. *Science of the Total Environment* 905C, 167118. DOI: [10.1016/j.scitotenv.2023.167118](https://doi.org/10.1016/j.scitotenv.2023.167118)

5. **Bodrud-Doza, M.**, Shammi, M., Bahlman, L., Islam, A.R.M.T., Rahman, M.M., 2020. Psychosocial and Socio-Economic Crisis in Bangladesh Due to COVID-19 Pandemic: A Perception-Based Assessment. *Front. Public Heal.* 8. DOI: [10.3389/fpubh.2020.00341](https://doi.org/10.3389/fpubh.2020.00341)
6. **Bodrud-Doza, M.**, Islam, S.M.D.U., Rume, T., Quraishi, S.B., Rahman, M.S., Bhuiyan, M.A.H., 2020. Groundwater quality and human health risk assessment for safe and sustainable water supply of Dhaka City dwellers in Bangladesh. *Groundw. Sustain. Dev.* 10, 100374. DOI: [10.1016/j.gsd.2020.100374](https://doi.org/10.1016/j.gsd.2020.100374)
7. **Bodrud-Doza, M.**, Bhuiyan, M.A.H., Islam, S.M.D.U., Quraishi, S.B., Muhib, M.I., Rakib, M.A., Rahman, M.S., 2019. Delineation of trace metals contamination in groundwater using geostatistical techniques: A study on Dhaka City of Bangladesh. *Groundw. Sustain. Dev.* 9. DOI: [10.1016/j.gsd.2019.03.006](https://doi.org/10.1016/j.gsd.2019.03.006)
8. **Bodrud-Doza, M.**, Bhuiyan, M.A.H., Islam, S.M.D.U., Rahman, M.S., Haque, M.M., Fatema, K.J., Ahmed, N., Rakib, M.A., Rahman, M.A., 2019. Hydrogeochemical investigation of groundwater in Dhaka City of Bangladesh using GIS and multivariate statistical techniques. *Groundw. Sustain. Dev.* 8, 226-244. DOI: [10.1016/j.gsd.2018.11.008](https://doi.org/10.1016/j.gsd.2018.11.008)
9. **Bodrud-Doza, M.**, Islam, S.M.D.U., Hasan, M.T., Alam, F., Haque, M.M., Rakib, M.A., Asad, M.A., Rahman, M.A., 2019. Groundwater pollution by trace metals and human health risk assessment in central west part of Bangladesh. *Groundw. Sustain. Dev.* 9, 100219. DOI: [10.1016/j.gsd.2019.100219](https://doi.org/10.1016/j.gsd.2019.100219)
10. **Bodrud-Doza, M.**, Islam, A.R.M.T., Ahmed, F., Das, S., Saha, N., Rahman, M.S., 2016. Characterization of groundwater quality using water evaluation indices, multivariate statistics and geostatistics in central Bangladesh. *Water Sci.* 30, 19-40. DOI: [10.1016/j.wsj.2016.05.001](https://doi.org/10.1016/j.wsj.2016.05.001)

Co-Authored Articles

1. Zihad S.M.R.A., Islam A.R.M.T., Siddique M.A.B., Mia M.Y., Islam M.S., Islam M.A., Bari A.B.M.M., **Bodrud-Doza M.**, Yakout S.M., Senapathi V., Chatterjee S. (2023). Fuzzy logic, geostatistics, and multiple linear models to evaluate irrigation metrics and their influencing factors in a drought-prone agricultural region. *Environ Res.* Volume 234, 116509. DOI: [10.1016/j.envres.2023.116509](https://doi.org/10.1016/j.envres.2023.116509)
2. Ferdous, S., Chowdhury, F.N., Ali, M.L., **Bodrud-Doza, M.**, Rahman, M.M. 2022. Assessment of urban sanitation status and management gaps in a metropolitan city, Bangladesh: Potential challenges to achieve SDG 6. *Front. Water* 4:950887. DOI: [10.3389/frwa.2022.950887](https://doi.org/10.3389/frwa.2022.950887)
3. Zaman, T., Tahsin, K.T., Rousseau, R.S., Kamal, A.B., Khan, M.R., Huq, S., **Bodrud-Doza, M.** 2022. An overview of disaster risk reduction and anticipatory action in Bangladesh. *Front. Clim.* 4:944736. DOI: [10.3389/fclim.2022.944736](https://doi.org/10.3389/fclim.2022.944736)
4. Rakib M. A., Quraishi S. B., Newaz M. A., Sultana J., **Bodrud-Doza M.**, Rahman M. A., Patwary M. A., Bhuiyan M. A. H. 2022. Groundwater quality and human health risk assessment in selected coastal and floodplain areas of Bangladesh. *Journal of Contaminant Hydrology.* Volume 249, 104041. DOI: [10.1016/j.jconhyd.2022.104041](https://doi.org/10.1016/j.jconhyd.2022.104041)
5. Rahman, M.S., Parvez, S., Begum, B., Quraishi S., Choudhury T., Fatema K., Hosen M., **Bodrud-Doza M.**, Rahman L., Sattar M.A 2022. Chemometric appraisal of water quality for domestic and agricultural purposes: a case study from establishing Rooppur Nuclear Power Plant (NPP) area, Pabna District, Bangladesh. *Environ Sci Pollut Res.* DOI: [10.1007/s11356-022-19308-6](https://doi.org/10.1007/s11356-022-19308-6)
6. Sultan, M.B., Choudhury, T.R., Alam, M.N., **Bodrud-Doza, M.**, Rahman, M.M. 2022. Soil, dust, and leaf-based novel multi-sample approach for urban heavy metal contamination appraisals in a megacity, Dhaka, Bangladesh. *Environmental Advances.* Volume 7, 100154. DOI: [10.1016/j.envadv.2021.100154](https://doi.org/10.1016/j.envadv.2021.100154)
7. Shammi, M., Rahmana, M.M., Ali, M.L., Khan, A.S.M., Siddique, M.A.B, Ashaduzzaman M., **Bodrud-Doza M.**, Alam, G.M.M, Tareq, S.M. 2022. Application of short and rapid strategic environmental assessment (SEA) for biomedical waste management in Bangladesh. *Case Studies in Chemical and Environmental Engineering.* Volume 5, 100177. DOI: [10.1016/j.cscee.2021.100177](https://doi.org/10.1016/j.cscee.2021.100177)
8. Rahman, M.M., **Bodrud-Doza, M.**, Shammi, M., Islam, A.R.M.T., Moniruzzaman Khan, A.S., 2021. COVID-19 pandemic, dengue epidemic, and climate change vulnerability in Bangladesh: Scenario assessment for strategic management and policy implications. *Environ. Res.* 192, 110303. DOI: [10.1016/j.envres.2020.110303](https://doi.org/10.1016/j.envres.2020.110303)
9. Shammi, M., **Bodrud-Doza, M.**, Islam, A.R.M.T., Rahman, M.M., 2021. Strategic assessment of COVID-19 pandemic in Bangladesh: comparative lockdown scenario analysis, public perception, and management for sustainability. *Environ. Dev. Sustain.* 23, 6148-6191. DOI: [10.1007/s10668-020-00867-y](https://doi.org/10.1007/s10668-020-00867-y)
10. Bhuiyan, M.A.H., **Bodrud-Doza, M.**, Rakib, M.A., Saha, B.B., Islam, S.M.D.U., 2021. Appraisal of pollution scenario, sources and public health risk of harmful metals in mine water of Barapukuria coal mine industry in Bangladesh. *Environ. Sci. Pollut. Res.* 28, 22105-22122. DOI: [10.1007/s11356-020-11999-z](https://doi.org/10.1007/s11356-020-11999-z)
11. Hasan, S., Islam, M.A., **Bodrud-Doza, M.**, 2021. Crisis perception and consumption pattern during COVID-19: do demographic factors make differences? *Heliyon* 7, e07141. DOI: [10.1016/j.heliyon.2021.e07141](https://doi.org/10.1016/j.heliyon.2021.e07141)
12. Hassan, M.S., Bhuiyan, M.A.H., Tareq, F., **Bodrud-Doza, M.**, Tanu, S.M., Rabbani, K.A., 2021. Relationship between COVID-19 infection rates and air pollution, geo-meteorological, and social parameters. *Environ. Monit. Assess.* 193, 29. DOI: [10.1007/s10661-020-08810-4](https://doi.org/10.1007/s10661-020-08810-4)
13. Islam, A.R.M.T., Hasanuzzaman, M., Shammi, M., Salam, R., **Bodrud-Doza, M.**, Rahman, M.M., Mannan, M.A., Huq, S., 2021. Are meteorological factors enhancing COVID-19 transmission in Bangladesh? Novel findings from a compound Poisson generalized linear modeling approach. *Environ. Sci. Pollut. Res.* 28, 11245-11258. DOI: [10.1007/s11356-020-11273-2](https://doi.org/10.1007/s11356-020-11273-2)

14. Islam, S.M.D.U., Mondal, P.K., Ojong, N., **Bodrud-Doza, M.**, Siddique, M.A.B., Hossain, M., Mamun, M.A., 2021. Water, sanitation, hygiene and waste disposal practices as COVID-19 response strategy: insights from Bangladesh. *Environ. Dev. Sustain.* 23, 11953-11974. DOI: [10.1007/s10668-020-01151-9](https://doi.org/10.1007/s10668-020-01151-9)
15. Mamun, M.A., Sakib, N., Gozal, D., Bhuiyan, A.I., Hossain, S., **Bodrud-Doza, M.**, Al Mamun, F., Hosen, I., Safiq, M.B., Abdullah, A.H., Sarker, M.A., Rayhan, I., Sikder, M.T., Muhi, M., Lin, C.-Y., Griffiths, M.D., Pakpour, A.H., 2021. The COVID-19 pandemic and serious psychological consequences in Bangladesh: A population-based nationwide study. *J. Affect. Disord.* 279, 462-472. DOI: [10.1016/j.jad.2020.10.036](https://doi.org/10.1016/j.jad.2020.10.036)
16. Siddique, A.B., Islam, A.R.M.T., Hossain, M.S., Khan, R., Akbor, M. A., Hasanuzzaman, M., Sajid, M.W.M, Mia, M.Y., Mallick, J., Rahman, M.S., Rahman, M.M, **Bodrud-Doza, M.** 2021. Multivariate statistics and entropy theory for irrigation water quality and entropy-weighted index development in a subtropical urban river, Bangladesh. *Environ Sci Pollut Res.* DOI: [10.1007/s11356-021-16343-7](https://doi.org/10.1007/s11356-021-16343-7)
17. Islam, A.R.M.T., Kabir, M.M., Faruk, S., Al Jahin, J., **Bodrud-Doza, M.**, Didar-ul-Alam, M., Bahadur, N.M., Mohinuzzaman, M., Fatema, K.J., Safiur Rahman, M., Choudhury, T.R., 2021. Sustainable groundwater quality in southeast coastal Bangladesh: co-dispersions, sources, and probabilistic health risk assessment. *Environ. Dev. Sustain.* DOI: [10.1007/s10668-021-01447-4](https://doi.org/10.1007/s10668-021-01447-4)
18. Rakib, M.A., Sasaki, J., Matsuda, H., Quraishi, S.B., Mahmud, M.J., **Bodrud-Doza, M.**, Ullah, A.K.M.A., Fatema, K.J., Newaz, M.A., Bhuiyan, M.A.H., 2020. Groundwater salinization and associated co-contamination risk increase severe drinking water vulnerabilities in the southwestern coast of Bangladesh. *Chemosphere* 246, 125646. DOI: [10.1016/j.chemosphere.2019.125646](https://doi.org/10.1016/j.chemosphere.2019.125646)
19. Islam, S.M.D., **Bodrud-Doza, M.**, Khan, R.M., Haque, M.A., Mamun, M.A., 2020. Exploring COVID-19 stress and its factors in Bangladesh: A perception-based study. *Heliyon* 6, e04399. DOI: [10.1016/j.heliyon.2020.e04399](https://doi.org/10.1016/j.heliyon.2020.e04399)
20. Islam, S.M.D., Safiq, M.B., **Bodrud-Doza, M.**, Mamun, M.A., 2020. Perception and Attitudes Toward PPE-Related Waste Disposal Amid COVID-19 in Bangladesh: An Exploratory Study. *Front. Public Heal.* 8. DOI: [10.3389/fpubh.2020.592345](https://doi.org/10.3389/fpubh.2020.592345)
21. Mamun, M.A., **Bodrud-Doza, M.**, Griffiths, M.D., 2020. Hospital suicide due to non-treatment by healthcare staff fearing COVID-19 infection in Bangladesh? *Asian J. Psychiatr.* 54, 102295. DOI: [10.1016/j.ajp.2020.102295](https://doi.org/10.1016/j.ajp.2020.102295)
22. Rahman, Mostafizur M.M., **Bodrud-Doza, M.**, Griffiths, M.D., Mamun, M.A., 2020. Biomedical waste amid COVID-19: perspectives from Bangladesh. *Lancet Glob. Heal.* 8, e1262. DOI: [10.1016/S2214-109X\(20\)30349-1](https://doi.org/10.1016/S2214-109X(20)30349-1)
23. Shammi, M., **Bodrud-Doza, M.**, Islam, A.R.M.T., Rahman, M.M., 2020. COVID-19 pandemic, socioeconomic crisis and human stress in resource-limited settings: A case from Bangladesh. *Heliyon* 6, e04063. DOI: [10.1016/j.heliyon.2020.e04063](https://doi.org/10.1016/j.heliyon.2020.e04063)
24. Rahman, M.M., **Bodrud-Doza, M.**, Siddiqua, M.T., Zahid, A., Islam, A.R.M.T., 2020. Spatiotemporal distribution of fluoride in drinking water and associated probabilistic human health risk appraisal in the coastal region, Bangladesh. *Sci. Total Environ.* 724, 138316. DOI: [10.1016/j.scitotenv.2020.138316](https://doi.org/10.1016/j.scitotenv.2020.138316)
25. Saha, N., **Bodrud-Doza, M.**, Islam, A.R.M.T., Begum, B.A., Rahman, M.S., 2020. Hydrogeochemical evolution of shallow and deeper aquifers in central Bangladesh: arsenic mobilization process and health risk implications from the potable use of groundwater. *Environ. Earth Sci.* 79, 1-18. DOI: [10.1007/s12665-020-09228-4](https://doi.org/10.1007/s12665-020-09228-4)
26. Habib, M.A., Islam, A.R.M.T., **Bodrud-Doza, M.**, Mukta, F.A., Khan, R., Bakar Siddique, M.A., Phoungthong, K., Techato, K., 2020. Simultaneous appraisals of pathway and probable health risk associated with trace metals contamination in groundwater from Barapukuria coal basin, Bangladesh. *Chemosphere* 242, 125183. DOI: [10.1016/j.chemosphere.2019.125183](https://doi.org/10.1016/j.chemosphere.2019.125183)
27. Islam, A.R.M.T., Islam, H.M.T., Mia, M.U., Khan, R., Habib, M.A., **Bodrud-Doza, M.**, Siddique, M.A.B., Chu, R., 2020. Co-distribution, possible origins, status and potential health risk of trace elements in surface water sources from six major river basins, Bangladesh. *Chemosphere* 249. DOI: [10.1016/j.chemosphere.2020.126180](https://doi.org/10.1016/j.chemosphere.2020.126180)
28. Shammi, M., Sultana, A., Hasan, N., Mostafizur Rahman, M., Saiful Islam, M., **Bodrud-Doza, M.**, Khabir Uddin, M., 2020. Pesticide exposures towards health and environmental hazard in Bangladesh: A case study on farmers' perception. *J. Saudi Soc. Agric. Sci.* 19, 161-173. DOI: [10.1016/j.jssas.2018.08.005](https://doi.org/10.1016/j.jssas.2018.08.005)
29. Bhuiyan, M.A.H., Chandra Karmaker, S., **Bodrud-Doza, M.**, Rakib, M.A., Saha, B.B., 2021. Enrichment, sources and ecological risk mapping of heavy metals in agricultural soils of dhaka district employing SOM, PMF and GIS methods. *Chemosphere* 263, 128339. DOI: [10.1016/j.chemosphere.2020.128339](https://doi.org/10.1016/j.chemosphere.2020.128339)
30. Akbor, M.A., Rahman, M.M., **Bodrud-Doza, M.**, Haque, M.M., Siddique, M.A.B., Ahsan, M.A., Bondad, S.E.C., Uddin, M.K., 2020. Metal pollution in water and sediment of the Buriganga river, Bangladesh: An ecological risk perspective. *Desalin. Water Treat.* 193, 284-301. DOI: [10.5004/dwt.2020.25805](https://doi.org/10.5004/dwt.2020.25805)
31. Khan, R., Islam, M.S., Tareq, A.R.M., Naher, K., Islam, A.R.M.T., Habib, M.A., Siddique, M.A.B., Islam, M.A., Das, S., Rashid, M.B., Ullah, A.K.M.A., Miah, M.M.H., Masrura, S.U., **Bodrud-Doza, M.**, Sarker, M.R., Badruzzaman, A.B.M., 2020. Distribution, sources and ecological risk of trace elements and polycyclic aromatic hydrocarbons in sediments from a polluted urban river in central Bangladesh. *Environ. Nanotechnology, Monit. Manag.* 14, 100318. DOI: [10.1016/j.enmm.2020.100318](https://doi.org/10.1016/j.enmm.2020.100318)
32. Azam, G., Huda, M.E., Bhuiyan, M.A.H., Mohinuzzaman, M., **Bodrud-Doza, M.**, Islam, S.M.D.-U., 2019. Climate Change and Natural Hazards Vulnerability of Char Land (Bar Land) Communities of Bangladesh: Application of the Livelihood Vulnerability Index (LVI). *Glob. Soc. Welf.* 8, 93-105. DOI: [10.1007/s40609-019-00148-1](https://doi.org/10.1007/s40609-019-00148-1)

33. Shammi, M., Rahman, M., Bondad, S., **Bodrud-Doza, M.**, 2019. Impacts of Salinity Intrusion in Community Health: A Review of Experiences on Drinking Water Sodium from Coastal Areas of Bangladesh. *Healthcare* 7, 50. [DOI: 10.3390/healthcare7010050](https://doi.org/10.3390/healthcare7010050)
34. Shammi, M., Rahman, M.M., Islam, M.A., **Bodrud-Doza, M.**, Zahid, A., Akter, Y., Quaiyum, S., Kurasaki, M., 2017. Spatio-temporal assessment and trend analysis of surface water salinity in the coastal region of Bangladesh. *Environ. Sci. Pollut. Res.* 24, 14273-14290. [DOI: 10.1007/s11356-017-8976-7](https://doi.org/10.1007/s11356-017-8976-7)
35. Rakib, M.A., Sasaki, J., Pal, S., Newaz, M.A., **Bodrud-Doza, M.**, Bhuiyan, M.A.H., 2019. An investigation of coastal vulnerability and internal consistency of local perceptions under climate change risk in the southwest part of Bangladesh. *J. Environ. Manage.* 231, 419-428. [DOI: 10.1016/j.jenvman.2018.10.054](https://doi.org/10.1016/j.jenvman.2018.10.054)
36. Ahmed, N., **Bodrud-Doza, M.**, Islam, A.R.M.T., Hossain, S., Moniruzzaman, M., Deb, N., Bhuiyan, M.A.Q., 2019. Appraising spatial variations of As, Fe, Mn and NO₃ contaminations associated health risks of drinking water from Surma basin, Bangladesh. *Chemosphere* 218, 726-740. [DOI: 10.1016/j.chemosphere.2018.11.104](https://doi.org/10.1016/j.chemosphere.2018.11.104)
37. Ahmed, N., **Bodrud-Doza, M.**, Islam, S.M.D.-U., Choudhry, M.A., Muhib, M.I., Zahid, A., Hossain, S., Moniruzzaman, M., Deb, N., Bhuiyan, M.A.Q., 2019. Hydrogeochemical evaluation and statistical analysis of groundwater of Sylhet, north-eastern Bangladesh. *Acta Geochim.* 38, 440-455. [DOI: 10.1007/s11631-018-0303-6](https://doi.org/10.1007/s11631-018-0303-6)
38. Islam, A.R.M.T., **Bodrud-Doza, M.**, Rahman, M.S., Amin, S.B., Chu, R., Al Mamun, H., 2019. Sources of trace elements identification in drinking water of Rangpur district, Bangladesh and their potential health risk following multivariate techniques and Monte-Carlo simulation. *Groundw. Sustain. Dev.* 9, 100275. [DOI: 10.1016/j.gsd.2019.100275](https://doi.org/10.1016/j.gsd.2019.100275)
39. Rahman, M.M., Islam, M.A., **Bodrud-Doza, M.**, Muhib, M.I., Zahid, A., Shammi, M., Tareq, S.M., Kurasaki, M., 2018. Spatio-Temporal Assessment of Groundwater Quality and Human Health Risk: A Case Study in Gopalganj, Bangladesh. *Expo. Heal.* 10, 167-188. [DOI: 10.1007/s12403-017-0253-y](https://doi.org/10.1007/s12403-017-0253-y)
40. Islam, M.A., Rahman, M.M., **Bodrud-Doza, M.**, Muhib, M.I., Shammi, M., Zahid, A., Akter, Y., Kurasaki, M., 2018. A study of groundwater irrigation water quality in south-central Bangladesh: a geo-statistical model approach using GIS and multivariate statistics. *Acta Geochim.* 37, 193-214. [DOI: 10.1007/s11631-017-0201-3](https://doi.org/10.1007/s11631-017-0201-3)
41. Rakib, M.A., Islam, S., Nikolaos, I., **Bodrud-Doza, M.**, Bhuiyan, M.A.H., 2017. Flood vulnerability, local perception and gender role judgment using multivariate analysis: A problem-based "participatory action to Future Skill Management" to cope with flood impacts. *Weather Clim. Extrem.* 18, 29-43. [DOI: 10.1016/j.wace.2017.10.002](https://doi.org/10.1016/j.wace.2017.10.002)
42. Islam, A.R.M.T., Shen, S.H., **Bodrud-Doza, M.**, 2017. Assessment of arsenic health risk and source apportionment of groundwater pollutants using multivariate statistical techniques in Chapai-Nawabganj district, Bangladesh. *J. Geol. Soc. India* 90, 239-248. [DOI: 10.1007/s12594-017-0705-9](https://doi.org/10.1007/s12594-017-0705-9)
43. Islam, A.R.M.T., Shen, S., Haque, M.A., **Bodrud-Doza, M.**, Maw, K.W., Habib, M.A., 2018. Assessing groundwater quality and its sustainability in Joypurhat district of Bangladesh using GIS and multivariate statistical approaches. *Environ. Dev. Sustain.* 20, 1935-1959. [DOI: 10.1007/s10668-017-9971-3](https://doi.org/10.1007/s10668-017-9971-3)
44. Atikul Islam, M., Zahid, A., Rahman, M.M., Rahman, M.S., Islam, M.J., Akter, Y., Shammi, M., **Bodrud-Doza, M.**, Roy, B., 2017. Investigation of Groundwater Quality and Its Suitability for Drinking and Agricultural Use in the South Central Part of the Coastal Region in Bangladesh. *Expo. Heal.* 9, 27-41. [DOI: 10.1007/s12403-016-0220-z](https://doi.org/10.1007/s12403-016-0220-z)
45. Islam, A.R.M.T., Ahmed, N., **Bodrud-Doza, M.**, Chu, R., 2017. Characterizing groundwater quality ranks for drinking purposes in Sylhet district, Bangladesh, using entropy method, spatial autocorrelation index, and geostatistics. *Environ. Sci. Pollut. Res.* 24, 26350-26374. [DOI: 10.1007/s11356-017-0254-1](https://doi.org/10.1007/s11356-017-0254-1)
46. Islam, A.R.M.T., Shen, S., **Bodrud-Doza, M.**, Rahman, M.S., 2017. Assessing irrigation water quality in Faridpur district of Bangladesh using several indices and statistical approaches. *Arab J Geosci* 10, 418. [DOI: 10.1007/s12517-017-3199-2](https://doi.org/10.1007/s12517-017-3199-2)
47. Safiur Rahman, M., Saha, N., Islam, A.R.M.T., Shen, S., **Bodrud-Doza, M.**, 2017. Evaluation of Water Quality for Sustainable Agriculture in Bangladesh. *Water, Air, Soil Pollut.* 228, 385. [DOI: 10.1007/s11270-017-3543-x](https://doi.org/10.1007/s11270-017-3543-x)
48. Islam, A.R.M.T., Shen, S., **Bodrud-Doza, M.**, Atiqur Rahman, M., Das, S., 2017. Assessment of trace elements of groundwater and their spatial distribution in Rangpur district, Bangladesh. *Arab. J. Geosci.* 10, 95. [DOI: 10.1007/s12517-017-2886-3](https://doi.org/10.1007/s12517-017-2886-3)
49. Bhuiyan, M.A.H., **Bodrud-Doza, M.**, Islam, A.R.M.T., Rakib, M.A., Rahman, M.S., Ramanathan, A.L., 2016. Assessment of groundwater quality of Lakshimpur district of Bangladesh using water quality indices, geostatistical methods, and multivariate analysis. *Environ. Earth Sci.* 75, 1020. [DOI: 10.1007/s12665-016-5823-y](https://doi.org/10.1007/s12665-016-5823-y)
50. Shammi, M., Rahman, R., Rahman, M.M., Moniruzzaman, M., **Bodrud-Doza, M.**, Karmakar, B., Uddin, M.K., 2016. Assessment of salinity hazard in existing water resources for irrigation and potentiality of conjunctive uses: a case report from Gopalganj District, Bangladesh. *Sustain. Water Resour. Manag.* 2, 369-378. [DOI: 10.1007/s40899-016-0064-5](https://doi.org/10.1007/s40899-016-0064-5)

OTHER SCHOLARLY OUTPUTS AND KNOWLEDGE MOBILISATION

Book Chapters

1. Sifa, S.F., Sultana, R., **Bodrud-Doza, M.**, 2021. Climate Change and COVID-19: Crisis Within Crises for Eradication of Poverty in Bangladesh. Muthu S.S (Eds), *COVID19: Environmental Sustainability and Sustainable Development Goals* (pp. 169-188). [DOI: 10.1007/978-981-16-3860-2_7](https://doi.org/10.1007/978-981-16-3860-2_7)

2. Ali, M.M., Afrin, S., Siddique, A.B., **Bodrud-Doza, M.**, Rahman M.M. (2022). Chapter 8 - Metal contamination in groundwater of Bangladesh. Tiwari, A.K., Kumar, A., Singh B.K., Singh, T.N., Suozzi, N., Matta, G., Russo, S.L. (Eds.) Water Scarcity, Contamination and Management (Volume 5, pp 109-131). DOI: [10.1016/B978-0-323-85378-1.00008-8](https://doi.org/10.1016/B978-0-323-85378-1.00008-8)

Working Papers

1. Coger, T., A. Dinshaw, S. Tye, B. Kratzer, M. Thazin Aung, E. Cunningham, C. Ramkissoon, S. Gupta, **Bodrud-Doza M.**, A. Karamallis, S. Mbewe, A. Granderson, G. Dolcemascolo, A. Tewary, A. Mirza, and A. Carthy. 2022. "Locally Led Adaptation: From Principles to Practice." Working Paper. Washington, DC: World Resources Institute (WRI). DOI: [10.46830/wriwp.21.00142](https://doi.org/10.46830/wriwp.21.00142)
2. Steinbach, D., Bahadur, A., Shakya, C., Thazin Aung, M., Burton, C., Gallagher, C., Mbewe, S., Greene, S., Regmi, B., Granderson, A., Ramkissoon, C., Kostka, W., Andon, L., Greenstone-Alefaio, T., Dolcemascolo, G., Gupta, S., Tewary, A., Lopez, M., Barnes, J., Mirza, A., **Bodrud-Doza, M.**, Akhter, F., Rozario, S., Reyes, C. (2022). The good climate finance guide for investing in locally led adaptation. IIED, London. [Read publication](#)

Conference Contributions

1. **Bodrud-Doza, M.** (2025). Towards implementing precision conservation practices in agricultural watersheds: A review of the use and prospects of Spatial Decision Support Tools and Systems. 1st International Conference on Canadian Agri-food & Rural Advisory, Extension, and Education (CAREE). Guelph, Ontario, Canada.
2. **Bodrud-Doza M.**, Rahman S., Bhuiyan M.A.H., Quraishi S. B., Choudhury T. R., Fatema K. J., Ji H. G. (2018). Health risk assessment of groundwater from shallow, intermediate and deep aquifer of Dhaka City, Bangladesh. 39th Annual Conference of Bangladesh Chemical Society. Bangladesh Chemical Congress-2018.
3. **Bodrud-Doza M.**, Chowdhury N., Rahman M. S. (2017). Risk assessment of trihalomethanes (THMs) formation in the supplied water of Dhaka city, Bangladesh. Abstract Book. Symposium on Environmental Chemistry for Securing Water Quality. Bangladesh Journal of Scientific and Industrial Research. 52 (Special Issue).
4. Asad M. A., **Bodrud-Doza M.**, Das M. (2018). Agricultural adaptation strategies to climate change - induced salinity intrusion in Southwest coastal Bangladesh. International Conference on Global Warming and Climate Change 2018. International Institute of Knowledge Management. 04th - 05th October 2018 in Bangkok, Thailand.
5. Rakib M.A., Islam S., Nikolaos I., **Bodrud-Doza M.**, Bhuiyan M. A. H. (2018). Flood vulnerability, local perception and gender role judgment using multivariate analysis. The 4th Gobeshona Conference for Climate Research in Bangladesh. 08-11 January, 2018 Dhaka, Bangladesh.
6. **Bodrud-Doza M.**, Jolly Y. N., Begum B. A., Fatema K. J., Bhuiyan M. A. H., Quraishi S. B., Rahman M. S. (2017). An integrated approach for groundwater pollution evaluation of Dhaka City using multivariate statistical analysis, groundwater pollution indices, geostatistical modeling and human health risk assessment. National Conference on Physics-2017. Bangladesh Physical Society. Atomic Energy Centre, Dhaka.
7. Anny F. A., Kabir M. M., **Bodrud-Doza M.** (2015). Determination of surface water quality in Savar urban area, Dhaka, Bangladesh. Annual Botanical Conference-2015. Bangladesh Botanical Society.

Technical Reports and Policy Contributions

1. **Bodrud-Doza, M.**, Burton, E., Dahunsi, J. E., Livingston, J., Mengistu-Zelleke, N., & Pollet, S. (2023). Upstream evidence report 2023. Arrell Food Institute & Ontario Agricultural College, University of Guelph. [Read publication](#)
2. Farhad S. M., Hasan M., Rahman M. S., Rahman H. M. H., Raju M. R. H., **Bodrud-Doza M.**, Sarkar S. (2022). Low-cost climate resilient urban housing: BRAC's experiences. Published by BRAC. 47 pages.
3. Khan A. S. M., **Bodrud-Doza M.**, Khan R. M., Siddique M. A. B., Ashaduzzaman M. (2020). Mainstreaming Climate Change at BRAC. Published by BRAC. 64 pages.
4. Nasir N., Rahman M. F., **Bodrud-Doza M.**, Khan A. S. M., (2020). Climate Resilience Framework of BRAC. Published by BRAC. 48 pages.
5. Huq S., Haque S., **Bodrud-Doza M.**, Nasir N., Yusuf M., Huq S. (2018). Assessment of Socio-economic Impacts of Cross-Border Electricity Trade (CBET) in Bangladesh. Published by Integrated Research and Action for Development (IRADe), India. 52 pages.
6. Schulenburg A., Kamruzzaman M., **Bodrud-Doza M.**, Mahmud M., Khan R. A., Haque M.A., Islam N., Talukder M. B. A., Auvi P., Rahman M. F. (2017). Adaptation Technology in Bangladesh. Jointly published by CCDB, ICCCAD and Practical Action. 73 pages.

Policy Brief

1. Huq S., **Bodrud-Doza M.**, Tahsin K. T., Khan M. R. (2022). What does the IPCC Working Group II say about Bangladesh in its Sixth Assessment Report?. Policy Brief. Dhaka, Bangladesh: ICCCAD. [Read publication](#)

Selected Public Writing and Research Communication

1. **Bodrud-Doza, M.** (27 August 2026). More than just survival. Development and Cooperation. [Read publication](#)
2. **Bodrud-Doza, M.** (15 July 2026). Following the Flow: Exploring the Geography of Cleaner Water. University of Guelph. [Read publication](#)

3. Coger T., Tye S., Karamallis A., Mirza A., **Bodrud-Doza M.**, Cunningham E.M. (01 November 2022). Getting Locally Led Adaptation Right: Examples from Around the World. [Read publication](#)
4. **Bodrud-Doza M.**, Tahsin K. T., (11 October, 2022). The ways to reinvent climate change adaptation in Bangladesh and beyond. [Read publication](#)
5. Huq S., Rozario S. R., **Bodrud-Doza M.** (18 August 2022). Are Bangladesh's development measures leading to climate change readiness or maladaptation?. [Read publication](#)
6. **Bodrud-Doza M.**, Mahfuz M. (18 July 2022). What local NGOs must do to secure climate finance. [Read publication](#)
7. Tahsin K. T., **Bodrud-Doza M.** (July 2022). Effective adaptation for avoiding maladaptation. [Read publication](#)
8. **Bodrud-Doza M.** (10 June 2022). Boosting disaster preparedness in the Ganges Delta. D+C Development and Cooperation. [Read publication](#)
9. Mahfuz M., **Bodrud-Doza M.** (9 May 2022). It's time to popularise climate resilient housing. [Read publication](#)
10. **Bodrud-Doza M.**, Khan S. (24 April 2022). Addressing food security and climate change through regenerative agriculture. [Read publication](#)
11. Huq S., **Bodrud-Doza M.**, Tahsin K. T. (2022). Resilience and prosperity under a changing climate. [Read publication](#)
12. Huq S., **Bodrud-Doza M.**, Tahsin K. T., Khan M. R. (20 March 2022). What does the IPCC say about Bangladesh in its Sixth Assessment Report of Working Group II? [Read publication](#)

TEACHING EXPERIENCE

Graduate Teaching Assistant

September 2022 - April 2024

Department of Geography, Environment and Geomatics, University of Guelph | Guelph, Ontario, Canada

- Supported laboratories, seminars, student consultation, assessment, rubric review, grading, and exam invigilation across five undergraduate geography courses.
- Courses: GEOG*1350 Earth: Hazards and Global Change; GEOG*2000 Geomorphology; GEOG*3000 Fluvial Processes; GEOG*2480 Mapping and GIS; GEOG*3480 GIS and Spatial Analysis.
- Delivered hands-on instruction in geomorphology and GIS and provided one-on-one technical and academic support to students.

Teaching interests: GIS and geospatial analysis; environmental modelling; watershed science; agricultural sustainability; environmental decision support. Remote sensing and GeoAI are developing areas for future teaching.

PROFESSIONAL AND LEADERSHIP EXPERIENCE

Operation and Business Development Manager

January - August 2022

International Centre for Climate Change and Development (ICCCAD), Independent University, Bangladesh

- Coordinated ICCCAD's research portfolio and Locally Led Adaptation Program, including project implementation, resource mobilisation, partnerships, and capacity development.
- Guided research papers, policy briefs, project reports, stakeholder training, and climate-resilience proposal development.

Consultant - Project Proposal Writer

August - November 2022

CARE Bangladesh

- Co-developed climate-resilience and ecosystem-based adaptation proposals with multidisciplinary teams.
- Co-authored a successful ecosystem-based adaptation proposal that secured €35 million from FCDO.

Consultant

March - August 2022

Ispahani Agro Limited

- Conducted research and assessments on climate-resilient seeds, bio-pesticides, sustainable agricultural investment, ESG, financial inclusion, and smallholder digital services.
- Supported landscape assessment and development of a digital system prototype for smallholder services, supply-chain monitoring, and product traceability.

Manager, Climate Resilience and Development

April 2018 - January 2022

BRAC Climate Change Program

- Led climate-adaptation project development and implementation, environmental and social safeguards, climate-smart agriculture, partnership building, and institutional climate-resilience frameworks.
- Worked with UNDP, government agencies, NGOs, and international donors; produced policy products, GIS outputs, training materials, and national reports connecting research with implementation.

Research Officer

October 2016 - March 2018

International Centre for Climate Change and Development (ICCCAD)

- Conducted climate-adaptation and resilience research, GIS analysis, technical writing, policy research, proposal development, and capacity building.
- Contributed to research on trapped populations, GCF feasibility work, national research planning, cross-border electricity trading, and adaptation technologies.

Junior Consultant

October - December 2016

International Union for Conservation of Nature (IUCN), Bangladesh

- Conducted GIS analysis and mapping to support wildlife conservation planning and Vulture Safe Zone initiatives.

ACADEMIC AND PROFESSIONAL SERVICE

- Peer reviewer for environmental science, water-resource, and sustainability journals, including Journal of Environmental Management, Environmental Research, Journal of Cleaner Production, Journal of Hydrology, Water Research, Agricultural Water Management, and Journal of Hazardous Materials.
- Committee Member, Arrell Food Chair in Food and Society Search Committee, University of Guelph (2023); represented graduate students in faculty hiring review and interviews.
- Member, Canadian Water Resources Association (2025 - present).
- Member, Canadian Geophysical Union (2024 - present).
- Member, Canadian Geomorphology Research Group (2024 - present).

TECHNICAL AND METHODOLOGICAL EXPERTISE

Programming & data science: Python, R, SQL, JavaScript; statistical analysis and machine learning for environmental and geospatial applications.

GIS & Earth observation: ArcGIS/ArcGIS Pro, QGIS, Google Earth Engine, ERDAS IMAGINE, WhiteboxTools, RockWorks, Surfer; terrain and raster analysis.

Watershed modelling & conservation planning: IMWEBs, SWAT/ArcSWAT, SWAT-CUP, Agricultural Conservation Planning Framework (ACPF), scenario analysis, BMP evaluation.

Decision analysis & visualization: Spatial prioritisation and optimisation, multi-criteria decision analysis, SPSS, Tableau, DeltaGraph, geospatial visualisation and dashboard development.

Field & survey data collection: ODK Collect, KoBoToolbox, Google Forms; environmental monitoring and mobile data collection.

Languages: English and Bangla - fluent in reading, writing, speaking, and listening.

SELECTED PROFESSIONAL DEVELOPMENT

- | | |
|---|---|
| • Spatial Data Science: The New Frontier in Analytics - Esri (2024) | • GIS for Climate Action - Esri (2024) |
| • Creating Python Scripts for Raster Analysis - Esri (2024) | • Remote Sensing Image Acquisition, Analysis and Applications - Coursera (2023) |
| • SWAT-CUP Calibration & Validation with ArcSWAT - Udemy (2024) | • Agricultural Conservation Planning Framework (ACPF) Technical Training - Iowa State University & USDA ACPF Hub (2022) |
| • Using LiDAR to Understand Ecosystems - Holland College (2024) | • Climate Smart Agriculture: Fundamentals; Data Management & Information Systems - Holland College (2024) |
| • Net-Zero Food Systems - Arrell Food Institute, University of Guelph (2024) | • Professional Proposal Writing, Fundraising & Donor Relations - BCTCI, Bangkok, Thailand (2019) |
| • Climate and Disaster Risk Financing - BRAC, BRAC University & World Bank Group (2019) | • Research Ethics (TCPS2: CORE 2022) - Government of Canada (renewed 2022) |